Sound and its representation

What is sound?

- Sound is caused by vibrations
- Vibrations create waves, travelling through a medium
- Humans perceive acoustic waves with their ears, as eardrum are vibrating, converting the signal for the brain
- It is usually represented as a sine wave, however, it is a longitudinal wave (compression/rarefaction) in air and water and a transversal wave in solids.

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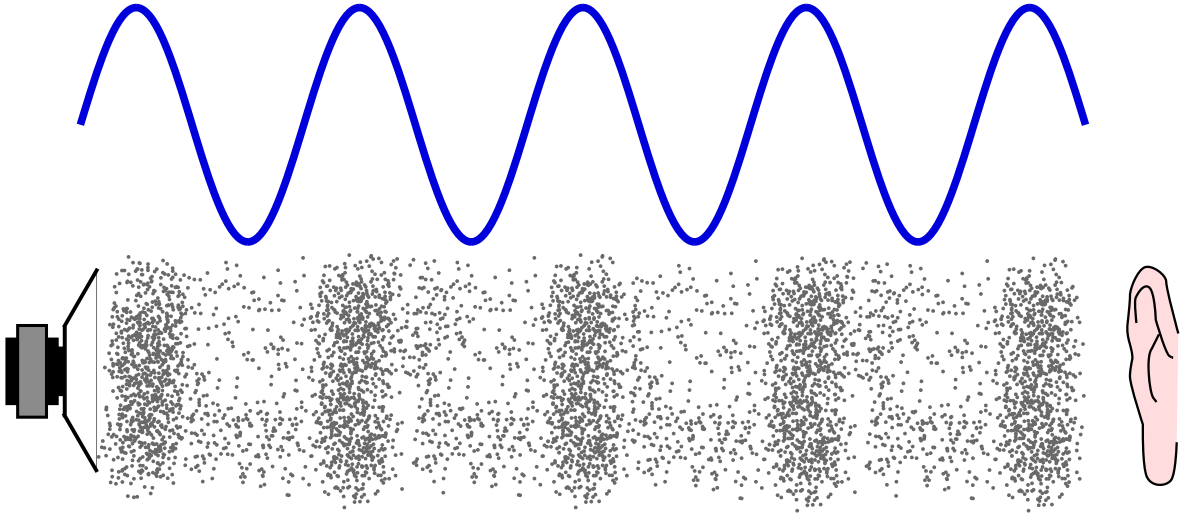
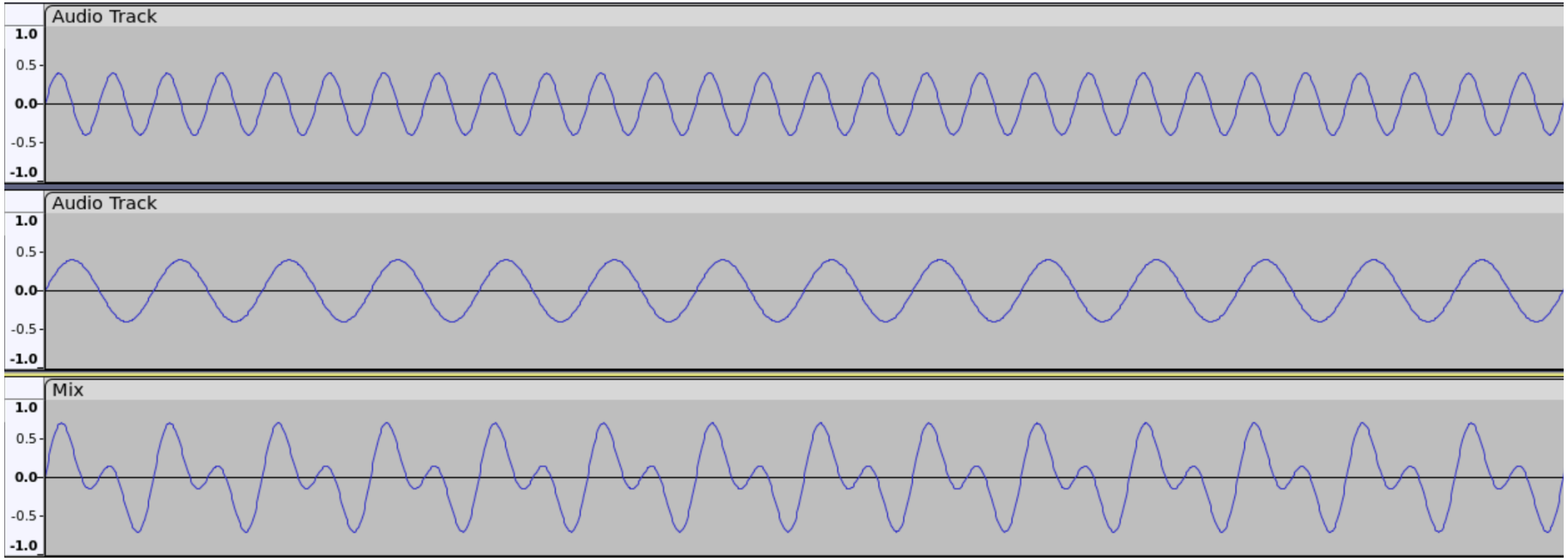
What is sound?



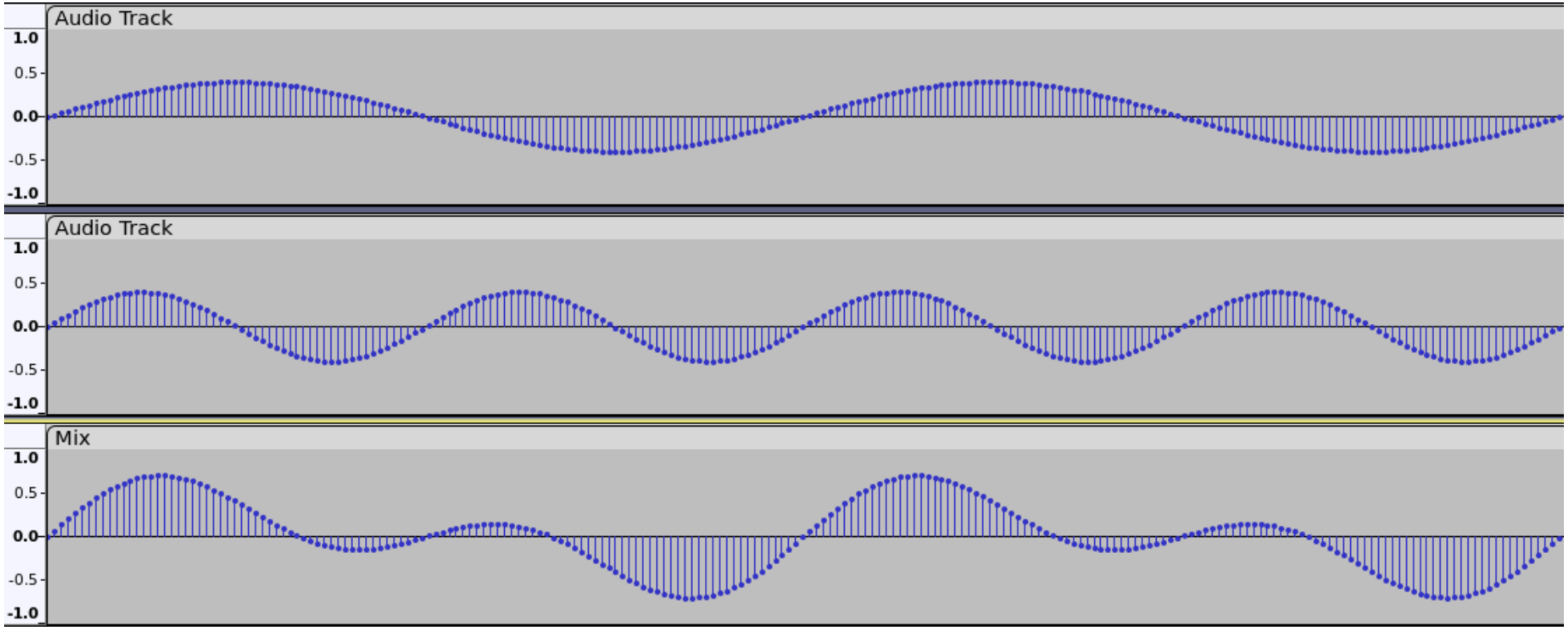
Sound characteristics

- Sound waves have a frequency, measured in Hertz (Hz), this is the pitch of the sound.
- They also have an amplitude, measured in decibels (dB), this is the loudness of the sound.
- Multiple waves of different frequencies and amplitude combine to create the actual sound with different qualities and timbre.



Sound digitization - samples

- Sound waves are continuous curves composed of a infinite number of points.
- For any point on the curve, it is possible to measure the audio level of this point.
- This is a sample. We can then take samples at regular interval to have a digital representation of the curve.



Sound digitization - sample rate

- The sample rate, or sampling frequency is the number of samples taken per seconds.
- If the sampling frequency is too slow, we may have aliasing issues were the sampled signal doesn't match the analog signal.
- The **Shannon-Nyquist theorem** states that the sampling frequency needs to be at least twice the maximum signal frequency to accurately digitize a signal.
- The Human ear can hear sound frequencies between approximately 20 Hz and 20 kHz.

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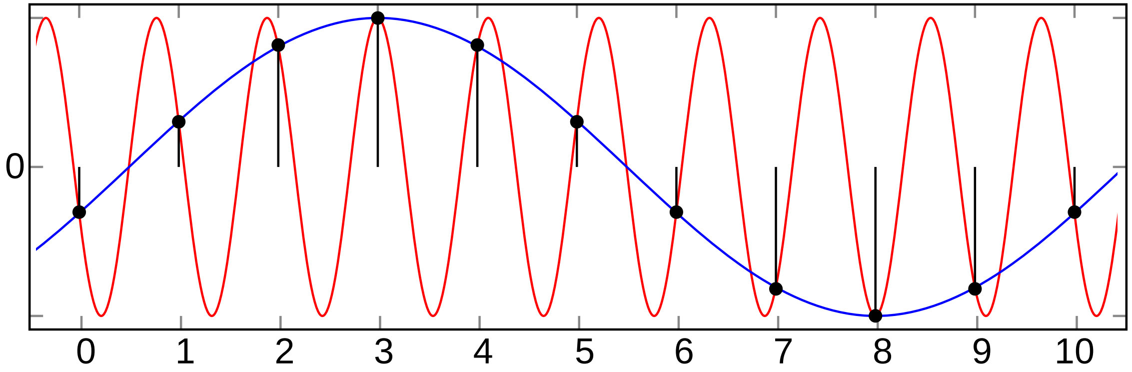

Sound digitization - sample rate

Aliasing example, the sampled signal is in blue

Sound digitization - sample size

- The sample value varies from 0 to the maximum amplitude value.
- If the amplitude is 1.0, then it varies from -1.0 and 1.0
- The sample size, in bits, then defines the resolution.
- Common sample sizes are 16 and 24 bits.
- 8 bits is getting very rare due to the poor audio quality and 32 bits samples can be used when specific alignment is required.

Sound digitization - sample format There are multiple ways to

store samples in memory or on disk:

- as signed integers
- as unsigned integers
- as floating points

Also, they can be stored in little-endian or big-endian order. For 24bit samples, packing can also differ: either they are packed on 3 bytes or they can be packed in a 32bit integer with the most significant byte being ignored.

Sound digitization - conclusions

- We can then store sound as a sequence of samples and the specific sample rate that was used.
- This method is called Linear Pulse-code modulation or LPCM.
- A sampling rate of about 40kHz is needed.

Sound digitization - example WAV WAV is a format based on RIFF

and has the following header:

Position	Value	Description
1 - 4	"RIFF"	RIFF FOURCC code
5 - 8		File size in bytes, minus 8 (32-bit integer).
9 –12	"WAVE"	WAVE FOURCC code
13-16	"fmt "	Format chunk marker (includes trailing space)
17-20	16	Length of format data, 16 for PCM
21-22	1	Audio format, 1 for PCM
23-24	2	Number of channels
25-28	48000	Sample rate
29-32	176400	Byte rate = (Sample rate **BitsPerSample** channels) / 8.
33-34	4	BlockAlign = (BitsPerSample*Channels) / 8
35-36	16	Bits per sample
37-40	"data"	Data chunk header

Sound digitization - example WAV WAV is a format based on RIFF

41-44		Size of the data section in bytes

Sound digitization - example WAV

```
0000000  52 49 46 46 24 EE 02 00 57 41 56 45 66 6D 74 20   RIFF$...WAVEfmt.
0000010  10 00 00 00 01 00 02 00 80 BB 00 00 00 EE 02 00   ................
0000020  04 00 10 00 64 61 74 61 00 EE 02 00 FF FF 00 00   ....data........
0000030  F5 02 00 00 E0 05 01 00 D0 08 FF FF AE 0B 00 00   ................
0000040  8E 0E 01 00 55 11 FE FF 19 14 04 00 C1 16 FC FF   ....U...........
0000050  62 19 03 00 E0 1B FD FF 50 1E 03 00 A2 20 FE FF   b.......P.......
```